

# Sipan Bareyan

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## EDUCATION

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### Paris-Saclay University, Faculty of Sciences of Orsay

Orsay, France

*Double Bachelor's Degree in Computer Science and Mathematics*

*Sept. 2023 – June 2026*

- Relevant coursework: Probability, Integration, Differential Calculus and Optimization, Linear Algebra (I-II), Advanced Algorithms, Data Structures and Algorithms, Statistical Learning, Numerical Analysis, Modeling and Simulation

### TUMO Center for Creative Technologies

Yerevan, Armenia

*Technology and Design Program*

*2015 – 2019*

- Courses taken: Programming, Game Development, 3D Modeling, Web Development, Robotics

### Anania Shirakatsy Lyceum

Yerevan, Armenia

*Secondary Education*

*Sept. 2011 – May 2023*

Overall GPA: 3.945/4

## EXPERIENCE

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### Research Intern – Numerical Methods

May 2026 – Jun. 2026

*Université Paris-Saclay*

*Orsay, France*

- Investigated automatic differentiation techniques for nonlinear differential equation solvers.
- Compared methods for gradient computation in PDE-constrained optimization problems.
- Implemented spectral PDE solvers, adjoint methods, and custom automatic differentiation rules in Julia.
- [github.com/bareyan/PDE\\_adjoint\\_solver](https://github.com/bareyan/PDE_adjoint_solver)

### Student Assistant

September 2025 –

*Lumen Library – Paris-Saclay University*

*Gif-sur-Yvette, France*

- Part-time work within the university library

### Intern – LAHDAK Team

June 2025 – July 2025

*LISN (Interdisciplinary Laboratory of Digital Sciences)*

*Gif-sur-Yvette, France*

- Worked on heterogeneous knowledge graphs
- Prepared data for training machine learning models
- Developed a knowledge graph exploration tool with automatic data cleaning based on statistical methods
- <https://github.com/bareyan/rust-kg-explorer>

### Front-end Web Development Intern

April 2023 – May 2023

*Volo*

*Yerevan, Armenia*

- Developed dynamic and responsive web applications with React

### Freelance Developer (Web & Mobile)

August 2020 – August 2023

*Remote*

- Developed websites and web applications for startups and companies.
- Developed mobile applications with Dart/Flutter for international clients.
- Kwork.ru profile: [https://kwork.ru/user/sb\\_dev](https://kwork.ru/user/sb_dev)

## PROJECTS

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### **EduViz** | (*in development*)

Devpost

- Prototype that transforms documents/notes into visual explanations (videos) with an educational structure.
- Examples: [youtube.com/playlist?list=PLu00ypjWn09trvhdCKQJ0c-5pb90uda-9](https://youtube.com/playlist?list=PLu00ypjWn09trvhdCKQJ0c-5pb90uda-9)

### **Rust KG Explorer**

[github.com/bareyan/rust-kg-explorer](https://github.com/bareyan/rust-kg-explorer)

- Knowledge graph explorer: entity navigation and SPARQL queries through a local interface.
- Import of datasets (Web Data Commons) or RDF files; tool used during my internship at LISN.
- Automatic cleaning and processing of heterogeneous graphs.

### **Tetris AI (university project)**

Report

- Implementation of Tetris and study of polyominoes (transformations, classes, generation) documented in a report.
- Heuristic AI with weights learned through a genetic algorithm (3,000–10,000 reported lines).

### **LaNoteX** | (*in development*)

LinkedIn Post

- Student application for organizing courses and generating study materials from imported content.
- Example repository: [github.com/bareyan/lecture\\_notes](https://github.com/bareyan/lecture_notes).

## PARTICIPATION IN COMPETITIONS AND EVENTS

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JFP 15 (15th Paris Region Programming Day) – 2nd place, Paris, France

Yeghiazaryan Cup 2021 & 2022 – Programming competition, Yerevan, Armenia

Independence Cup 2021 – Programming competition, Yerevan, Armenia (special prize)

Robotex 2019 – International robotics competition, Yerevan, Armenia (smart city robot)

Digitech Expo 2019 – Yerevan, Armenia (presentation of a mobile drone project)

## VOLUNTEERING

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Volunteer at Anania Shirakatsy Lyceum, Yerevan, Armenia — Jan. 2019 – May 2023

Volunteer at CEPFA, Yerevan, Armenia

## TECHNICAL SKILLS

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**Programming Languages** : Python, C, C++, Java, Rust, JavaScript, Dart, OCaml, Processing

**Query Languages** : SQL, SPARQL

**Frameworks** : React, Flutter

**Tools** : Linux, Git, Blender

**Markup Languages and Data Formats** : LaTeX, HTML & CSS, RDF, RDFa

**Libraries** : NumPy, Matplotlib, PyTorch, SciPy, Scikit-learn

## LANGUAGES

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Armenian (native), English (advanced – C1), French (professional proficiency), Russian (advanced)

## STANDARDIZED TESTS

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TOEIC (March 2026): 990 — SAT (March 2022) Mathematics: 780 — DELF B2 obtained (November 2022)